# Exit Ticket · reflect and retrieve

## Exit Ticket

When increasing an exponent by 1 during alignment, the mantissa must shift right 1 place to preserve its value.

## Use key vocabulary from memory

1. Explain exponent alignment in your own words.

2. Explain rounding in your own words.

3. Explain truncation in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Floating-point arithmetic and precision

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Normalise positive $0.0100_2 \times 2^4$.

### Q2

What is $-1$ in 8-bit two's complement?

### Q3

Calculate $0.5 \times 2^4$.

### Q4

Calculate binary $0.1000_2 + 0.0100_2$.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Align exponents — Before adding mantissas, both terms need the same power of two. Shift B right one place when raising its exponent by one.

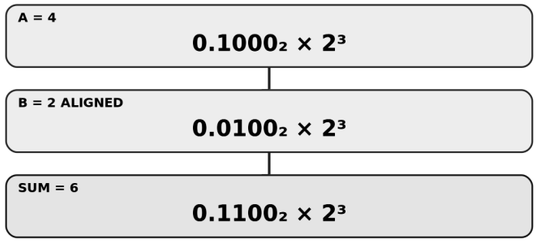
## Core idea 2

Add the mantissas — $0.1000_2 + 0.0100_2 = 0.1100_2$; keep the common exponent 3.

## Core idea 3

Check in decimal — $0.1100_2 \times 2^3 = 0.75 \times 8 = 6$, matching 4 + 2.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Explain how exponent alignment can lose precision when the mantissa has a fixed number of bits.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Why must exponents be aligned before binary floating-point addition?

## Question 2

Calculate and normalise $0.1010 \times 2^3 - 0.1000 \times 2^2$. Retain all shown mantissa bits during working.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

exponent alignment — Express operands using the same power of two.

rounding — Choose a nearby representable value for an exact result.

truncation — Discard bits beyond the available precision.

overflow — Numerical result exceeds the representable range.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

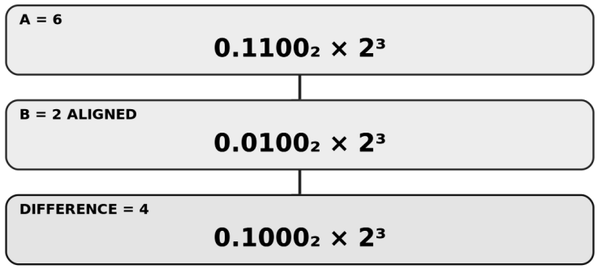
## Problem

Calculate $0.1100_2 \times 2^3$ minus $0.1000_2 \times 2^2$.

## Follow the worked method

1. Align — B=$0.1000 \times 2^2$ becomes $0.0100 \times 2^3$.
2. Subtract — $0.1100 - 0.0100 = 0.1000$.
3. Check — $0.1000_2 \times 2^3 = 4$; this matches decimal $6 - 2$.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Align
Raise B's exponent 1→3 by shifting its mantissa right twice: 0.1000 becomes 0.0010.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.